ISABELLE FERNAN

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Profile

Final-year Bachelor of Information Technology (Software Development) student at Monash University with industry experience in machine learning, computer vision, and software development. Experience in developing full-stack web applications, backend systems, and mobile applications built through industry projects, professional education, and client-facing roles with a strong foundation in problem solving, technical, leadership and communication skills.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, PHP, C++, Swift, Kotlin

Frameworks and Development: React, Angular, Node.js, Express.js, CakePHP, SwiftUI, FastAPI, HTML/CSS, Tailwind CSS, Bootstrap, REST APIs

Databases and Infrastructure: MySQL, SQLite, MongoDB, Redis, Docker, Git, GitHub, Linux, Nix, Google Cloud/GKE, AWS fundamentals

AI and Computer Vision: PyTorch, Detectron2, object detection evaluation, bounding-box analysis, optical flow, and video evaluation; MLflow and ClearML

Other Tools and Practices: Postman, VS Code, Xcode, Android Studio, Figma, Agile, UI/UX design, technical troubleshooting

Professional Experience

Silverpond Melbourne CBD, VIC
Machine Learning Intern 2026–Present

- Evaluated industrial pallet object-detection predictions against ground-truth bounding boxes, investigating prediction mismatches, annotation errors, and cases where incorrect labels distorted model assessment.
- Reviewed vision-language model (VLM) hazard classifications and documented discrepancies between model output categories and ground-truth labels rather than assuming direct class equivalence.
- Analysed optical-flow-based motion graphs from poultry video across activity levels, age groups, and recording scenarios to assess how well motion signals represented observed behaviour.
- Prepared for a Detectron2 object-detection training workflow by reviewing Python code, training metadata, dataset schemas, and class-index mappings; investigated MLflow/ClearML workflows before dataset access was available.

Skills Samurai Elsternwick and Glen Waverley, VIC
Coding and Maths Coach 2025–Present

- Teach Python, Java, Scratch, and Lua to students aged 6–16, adapting technical explanations and hands-on exercises to different learning levels.
- Communicate student progress with parents and troubleshoot classroom computers, software, networking, and 3D printers.

Monash Nexus for Emerging Technologies (MNET) Clayton, VIC
Co-CEO / Technology Organisation Leader 2024–2025

- Co-led an 80+ member emerging-technology organisation, coordinating technical events, multidisciplinary teams, and engagement with industry contacts.
- Contributed to the design and development of a modern website for a student technology organisation which included UI/UX design and frontend implementation using component-based architecture.
- Organised and executed tech events and workshops, connecting students with industry partners.
- Contributed to team rebranding, improving visibility and engagement across platforms.

Education

Monash University Expected completion: 2026
Bachelor of Information Technology (Software Development)

Zuitt Intensive Coding Bootcamp 2023
Full Stack Web Development (MongoDB, Express, React, Node.js)

Selected Technical Projects

Moving House 24/7 – Business Management Web Platform More Info | Live Demo

CakePHP, PHP, MySQL, JavaScript, Bootstrap

- Developed customer-facing and administration interfaces for a team-built removal and storage management platform, including responsive pages and form workflows.
- Implemented a threaded customer inquiry/reply interface, role-based navigation and access validation, and UI for staff, job and resource allocation.
- Contributed to CMS content management, admin dashboard interfaces, data validation, and debugging across customer and staff workflows.

Secure Finance Group – Client Web Platform More Info

React, TypeScript, Vite, CSS Modules, Express

- Redesigned responsive customer and admin interfaces with reusable React components and shared CSS Modules for a finance brokerage client.
- Improved appointment-booking forms, dynamic time-slot layouts, and frontend routing; served as client liaison to clarify requirements and coordinate feedback.

SmartPark API – Parking Detection Service More Info

Python, FastAPI, React, Docker, GKE, Redis

- Worked on a containerised computer-vision API and React dashboard for parking-space analysis, with YOLO inference and endpoints for ranked car-park results.
- Integrated cloud deployment components including Google Kubernetes Engine, shared Redis state, and model access through Google Cloud Storage; tested health/readiness and API workflows.

Aussie EcoGuard – Wildlife Media Analysis (final development)

TypeScript, Node.js, SQLite, Computer Vision

- Implemented backend media-upload validation, SHA-256 deduplication, persistent SQLite metadata, thumbnail generation, and retrieval for a team project.
- Built species/count search, configurable animal detection, and media-management workflows, with automated backend tests; cloud deployment is separate from the local implementation.

Book Tracker – Native iOS App (final development)

Swift, SwiftUI, Firebase, Open Library API

- Built native reading-library workflows with book details, reading progress, session history, and reading statistics using SwiftUI.
- Integrated book search via the Open Library API and developed local data-management features; reading goals, custom lists, reminders and Firebase-related integration are undergoing final testing.

CIFAR-10 Image Classifier More Info | GitHub | Live Demo

Python, PyTorch, React, TypeScript, ONNX

- Trained and evaluated a convolutional neural network on the CIFAR-10 image dataset using PyTorch; used a reproducible Nix development environment.
- Developed a browser-based model demonstration with a React/TypeScript interface and ONNX export.

Additional Leadership and Communication

Esports Commentator and Event Host – Ardent (Tier One) and Freelance 2021–2024
Delivered analytical live commentary at 40+ esports events, including 15+ international events, translating complex, fast-changing scenarios for live audiences.

Founder / Cultural Affairs Officer – Filipino Association Monash 2025–2026
Helped establish a student organisation, coordinate events and community engagement.